

Effective Archimedean Stark Theorems over Real Quadratic Fields: Quadratic Support, Shintani Transfer, and CM Descent

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Abstract

We prove a uniform formula for every one-place real-quadratic Stark invariant whose differenced Fourier support is quadratic. The formula is an explicit product of relative units, with rational exponents computed from class numbers, imprimitive Euler factors, and an exact relative-regulator index. Beyond this quadratic floor, we translate Shintani’s index-two theorem into the one-place notation, give a general all-embeddings height-rigidity criterion, and use them to identify eight explicit packets with support orders up to ten. The examples comprise five order-six packets—including one at a ramified 3-power conductor—two quartic packets, and an order-ten packet. We also prove a cyclic-quartic Fourier-to-CM-norm theorem from Stark’s imaginary-quadratic rank-one theorem and apply it to five primitive quartic packets, including examples with $e = |\mu(E)| = 2, 6, 8$. The $e = 8$ route proves the packet over $\mathbb{Q}(\sqrt{6})$; a separate $e = 12$ computation is retained only as a cross-check because its auxiliary-prime valuation argument is not used in the proof. Two structural lemmas prove that an absolutely abelian one-place ray field has trivial differenced support and that nontrivial support forces even Shintani index. The latter is independently consistent with all 446 genuinely reconstructed odd-index cases in a separate census. Exact and Arb certificates accompany the written proofs; no unproved Stark conjecture or floating-point recognition enters a proof.

1 Introduction and claim boundary

Let K be real quadratic and let $\mathfrak{m} = \mathfrak{f}\infty_2$ be a ray modulus containing one real place. For $K = \mathbb{Q}(\sqrt{D})$ with $D > 0$, our convention is that ∞_2 is the negative-square-root embedding $\sqrt{D} \mapsto -\sqrt{D}$, which is the first real place in the pinned PARI ordering; ∞_1 is the positive-square-root embedding. For $A \in \text{Cl}_{\mathfrak{m}}(K)$, let R be the sign class and put

$$Z_{\mathfrak{m}}(s, A) = \zeta_{\mathfrak{m}}(s, A) - \zeta_{\mathfrak{m}}(s, RA), \quad X_A = \exp Z'_{\mathfrak{m}}(0, A). \quad (1)$$

The numbers X_A are positive real analytic invariants. We call the finite Artin-indexed tuple

$$\mathcal{X}_{\mathfrak{m}} = (X_A)_{A \in \text{Cl}_{\mathfrak{m}}(K)}$$

the *one-place packet*; repeated values may be suppressed when a smaller Artin orbit is displayed. The word “packet” carries no algebraicity assumption. Stark’s rank-one conjecture predicts algebraic units with the prescribed Artin action, but that conjecture is not known in this generality. The purpose of this paper is to identify explicit packets only where a proved theorem already supplies algebraicity and where the remaining analytic-to-algebraic comparison can be certified.

The word *unconditional* has this precise meaning throughout: each displayed identity follows from a published proved theorem under hypotheses checked exactly in the case bundle. It does not mean that the general real-quadratic rank-one Stark conjecture is proved. Shintani's theorem is used only in its index-two range [3]. The underlying rank-one conjecture for quadratic extensions is the classical proved case of Tate [5, Thm. IV.5.4]; the contribution of Engine A is the uniform one-place formula, including its imprimitive Euler factors, relative-regulator index, and exact per-character exponents. Arakawa's relative-index formula for quadratic base fields is closely related [6]. CM descent uses Stark's proved theorem for abelian extensions of imaginary quadratic fields [4]. Numerical PARI recognition, when retained as a cross-check, is outside the proof chain.

Terminology and quantitative conventions

Fourier inversion of (1) involves precisely the ray characters χ for which $\chi(R) = -1$. The *support order* is the maximum order of such a character. Labels such as RQ-000190 are stable locators in the accompanying corpus; they are not mathematical notation, and every use below also states the field and modulus.

If θ is a supported quartic character and $U = \ker \theta$, its *primitive quartic packet* is the geometric fiber mean

$$Y_{\bar{A}} = \left(\prod_{B \in U} X_{AB} \right)^{1/|U|}, \quad \bar{A} \in G/U. \quad (2)$$

It is a positive analytic invariant indexed by $G/U \simeq C_4$. This definition is essential: it removes every Fourier component that does not factor through the selected quartic quotient. When the full differenced support is the inverse pair $\{\theta, \bar{\theta}\}$, X_A is constant on U , and $Y_{\bar{A}} = X_A$.

For Engine B, a *safe exponent* is an explicitly computed integer m for which Shintani's theorem proves that every X_A^m is algebraic in the asserted ray field. If the analytic and algebraic logarithms differ by balls of radius at most δ , the height comparison bounds the quotient by $m\delta$. The reported *margin* is

$$\mathcal{M} = \frac{V_{\min}}{m\delta}, \quad (3)$$

where $V_{\min}$ is the minimum applicable Voutier lower bound over the certified degree window. Thus $\mathcal{M} > 100$ is the pre-registered acceptance condition. A *cone evaluator* is the finite Shintani–Yamamoto decomposition of a partial zeta derivative into double-sine or Mellin-integral terms; all such terms below are evaluated as Arb balls [8, 7].

Fourier, character, and Artin conventions. We use

$$L_{\mathfrak{m}}(s, \chi) = \sum_{A \in G} \chi(A) \zeta_{\mathfrak{m}}(s, A), \quad \zeta_{\mathfrak{m}}(s, A) = \frac{1}{|G|} \sum_{\chi \in \hat{G}} \chi(A)^{-1} L_{\mathfrak{m}}(s, \chi).$$

The Artin map is normalized so that the class B sends the label A to AB . For a quartic quotient we choose $\theta(\bar{s}) = i$; on the imaginary-quadratic side we choose $\psi(\sigma) = i$, and the exact normal-closure identification sends $\bar{s}$ to σ . Replacing both generators by their inverses is the same convention with relabeled classes; replacing only one is forbidden and would interchange the labels $\bar{s}$ and $\bar{s}^3$.

How the results fit together

There are three theorem mechanisms. Engine A is a uniform closed-form formula for quadratic Fourier support. It is also the algorithmically closed stratum: Corollary 6 evaluates every quadratic-support packet by exact finite arithmetic in class groups and unit lattices of quartic

fields, with no analytic enclosure, safe exponent, or height comparison. Engines B and C are instead per-case certified-analysis mechanisms; see the cost discussion at the end of Section 3. Engine B reaches higher character orders when Shintani’s maximal absolutely abelian subfield has index two: Proposition 8 translates the algebraicity theorem and Lemma 9 turns certified archimedean comparisons into exact identities. The order-six cases on either side of ramification isolate the index/wildness condition, rather than character order, as the relevant obstruction. Engine C passes from a quartic quotient packet to an abelian cyclic-quartic extension of an imaginary quadratic field. Exact linear reinduction identifies the L -function, and Theorem 10 converts Stark’s proved unit into the Artin-labeled positive packet. A packet proved by both Engines B and C tests the overlap of the two theorem bases. Lemmas 11 and 12 explain two further structural boundaries.

Main theorems

Theorem 1 (Uniform quadratic-support theorem). *For every real quadratic K , every one-place modulus $\mathfrak{m} = \mathfrak{f}\infty_2$, and every packet whose differenced Fourier support consists only of quadratic characters, each X_A is an explicit rational power of a product of relative units. Corollary 6 gives one closed product formula whose exponents are expressed entirely in terms of exact class numbers, roots of unity, imprimitive Euler factors, and relative-regulator indices. No numerical approximation is required to state or prove the formula.*

Theorem 2 (Selected higher-order packets). *Use the integral basis $1, \omega$, where $\omega = (1 + \sqrt{D})/2$ for $D \equiv 1 \pmod{4}$ and $\omega = \sqrt{D}$ otherwise. The short modulus labels below are defined by exact HNF matrices in Appendix A. For every row below, the packet $\mathcal{X}_{\mathfrak{f}\infty_2}$ is the Artin-labeled set of positive roots of the relative polynomial printed in Section 4 or 6.*

case	K	modulus	support	m	margin
RQ-000190	$\mathbb{Q}(\sqrt{7})$	$\mathfrak{p}_7$	2, 6	4032	> 5688
RQ-000419	$\mathbb{Q}(\sqrt{14})$	$\mathfrak{p}_7$	2, 6	4032	> 7315
RQ-000108	$\mathbb{Q}(\sqrt{5})$	$\mathfrak{f}_{45}$	4	2880	> 2460
RQ-000021	$\mathbb{Q}(\sqrt{2})$	(7)	2, 6	2016	> 4261
RQ-002057	$\mathbb{Q}(\sqrt{57})$	$\mathfrak{f}_{27}$	2, 6	2592	> 748
RQ-002955	$\mathbb{Q}(\sqrt{77})$	$\mathfrak{p}_7$	2, 6	4032	> 5151
RQ-001107	$\mathbb{Q}(\sqrt{33})$	$\mathfrak{p}_{11}$	2, 10	15840	> 5817
RQ-000458	$\mathbb{Q}(\sqrt{14})$	$\mathfrak{f}_{72}$	4	1152	> 6470

Here m is the safe exponent of Proposition 8, and “margin” has the meaning in (3). Each row includes exact ray-field identification, unconditional class and unit groups, Sturm isolation, all-embeddings Arb bounds, and exact Artin labels.

Theorem 3 (Selected cyclic-quartic CM packets). *Use the same integral-basis convention as in Theorem 2. For every row below, the primitive quartic packet (2) is the Artin-labeled set of positive numbers whose polynomial is printed in Section 6.*

case	K	modulus	CM bases	e used in proof	polynomial
RQ-001280	$\mathbb{Q}(\sqrt{35})$	$\mathfrak{f}_{32}$	$\mathbb{Q}(\sqrt{-10}), \mathbb{Q}(\sqrt{-14})$	2, 2	P_{35}
RQ-001569	$\mathbb{Q}(\sqrt{42})$	$\mathfrak{f}_{12}$	$\mathbb{Q}(\sqrt{-7}), \mathbb{Q}(\sqrt{-6})$	6, 6	P_{42}
RQ-007519	$\mathbb{Q}(\sqrt{186})$	$\mathfrak{f}_{12}$	$\mathbb{Q}(\sqrt{-6}), \mathbb{Q}(\sqrt{-31})$	6, 6	P_{186}
RQ-001894	$\mathbb{Q}(\sqrt{51})$	$\mathfrak{f}_{75}$	$\mathbb{Q}(\sqrt{-15}), \mathbb{Q}(\sqrt{-85})$	6, 6	P_{51}
RQ-000129	$\mathbb{Q}(\sqrt{6})$	$\mathfrak{f}_8$	$\mathbb{Q}(\sqrt{-2})$	8	P_6

In each route the relative extension over the imaginary quadratic base is cyclic quartic, the exact character identity includes all imprimitive Euler factors, Stark’s global-unit hypothesis $|S| \geq 3$ holds, and the free unit supplied by Stark is certified to be an $e/2$ -th power modulo torsion. Exact common-normal-closure identities implement Theorem 10. For RQ-000129 only the $e = 8$ route is used in this theorem; the $e = 12$ auxiliary-prime calculation is not part of the proof.

Historical scope. Earlier papers in this program established the quadratic dimension-four calibration and an unconditional support-order-eight packet over $\mathbb{Q}(\sqrt{3})$ at $(5)\infty_2$, with safe exponent 5760 [1]. That order-eight packet is not repeated in the selected tables here; it is another specialization of Proposition 8 and Lemma 9. The second paper then used unconditional order-six one-place inputs over $\mathbb{Q}(\sqrt{2})$, including the ray-(7) stratum, in proving both dimension-seven strata [2]. Thus RQ-000021 is a streamlined effective reidentification of a previously used arithmetic input, not a new priority claim. The present work places those mechanisms in a common effective framework and supplies further order-six replications.

Relation to earlier index-formula results. At the quadratic layer, Tate’s Theorem IV.5.4 proves the full rank-one conjecture for quadratic extensions in this setting, including signs [5]; see also Arakawa’s index formula for quadratic base fields [6]. Thus the algebraicity underlying Engine A is classical. The new content of Theorem 1 is its uniform explicit expression in the one-place normalization through I_χ , E_χ , and the resulting exact per-character exponents.

Roblot [13, Thms. 6.1 and 7.1] proves unconditionally that the Stark-unit index formulae admit solutions for cyclic quartic extensions and, under additional class-number and wild-ramification hypotheses, for cyclic sextic extensions. The resulting comparison with L' -values is a weak Stark statement up to complex absolute values; it does not identify the individual Artin-labelled components of a one-place Shintani packet. We therefore make no claim to the first unconditional weak Stark result in degree four or six.

For the five order-six rows of Theorem 2, an exact audit finds that all five satisfy Roblot’s (A1)–(A3) and have ray-field class number $h_H = 1$. Theorem 7.1 of [13] applies to RQ-000190, RQ-000419, RQ-000021, and RQ-002955. It does not apply to RQ-002057: the prime above 3 has relative ramification index 6 in H/K , and is therefore wild. Supplementary Table S3 records the check. Roblot’s (A4) is not a hypothesis of these existence theorems; it is required in his earlier deduction that an assumed Stark unit satisfies the local index property (P2).

The contribution here is componentwise identification: the positive one-place invariants are proved to be the Artin-labelled roots of an explicit ray-field polynomial. This recovers the phase and labelling information not determined by the index formula alone. The order-ten case lies outside the quartic and sextic scope of [13]. We are not aware of an earlier unconditional, componentwise Artin-labelled identification of a one-place Shintani packet with support order ten. This survey-bounded historical observation plays no role in the proofs. These archimedean results are also distinct from Gross–Stark computations [15] and from the proved finite-place Brumer–Stark theory [14].

Theorems 2 and 3 are deliberately selected-results theorems, not census-completeness statements. Counts, conductor trends, and a complete frontier taxonomy require the separately audited census and will appear elsewhere. This separation prevents provisional classification data from entering a theorem proof. The prior numerical and conditional computations of Dummit–Sands–Tangedal and Cohen–Roblot remain prior work rather than claims displaced here [10, 11, 12]. More precisely, the 1997 Dummit–Sands–Tangedal computations concern totally real cubic bases and are numerical; their 2003 exponent-two theorems overlap the theorem class of Engine A, which is consequently not advertised as a historical first. Cohen–Roblot construct Hilbert class fields conditionally from Stark units and verify the resulting fields independently, whereas every promoted object here is a one-place ray field with nontrivial finite modulus. Thus these are mathematically adjacent precedents, not identical class-field objects. The remaining order-ten historical statement refers to the bounded literature comparison recorded in the supplement.

2 The quadratic-support theorem

Let $G = \text{Cl}_m(K)$, and suppose every character χ with $\chi(R) = -1$ has order two. Write χ° for the associated primitive Hecke character and L_χ/K for the quadratic extension it cuts out. The selected real place becomes complex in L_χ , while the other real place splits, so L_χ has signature $(2, 1)$.

Let ϵ_K generate $E_K/\mu(K)$. Modulo torsion, use the logarithmic embedding

$$\lambda_L(u) = (\log |\tau_1 u|, \log |\tau_2 u|, 2 \log |\tau_3 u|), \quad (4)$$

where τ_1, τ_2 are real and τ_3 is one of the complex embeddings. Let u_χ generate the primitive rank-one kernel of

$$N_{L_\chi/K} : E_{L_\chi}/\mu(L_\chi) \longrightarrow E_K/\mu(K),$$

and orient it by $\log |\tau_1 u_\chi| > 0$. In any exact basis of $E_{L_\chi}/\mu(L_\chi)$, write v_K and v_χ for the coordinate columns of ϵ_K and u_χ , and define

$$I_\chi = [E_{L_\chi}/\mu(L_\chi) : \langle E_K/\mu(K), u_\chi \rangle] = |\det(v_K, v_\chi)|. \quad (5)$$

Proposition 4 (Explicit quadratic formula). *With the Fourier, character, and Artin conventions fixed above, let E_χ be the exact product of imprimitive Euler factors at zero. Then*

$$L'_m(0, \chi) = E_\chi \frac{h_{L_\chi}}{h_K} \frac{w_K}{w_{L_\chi}} \frac{2}{I_\chi} \log |\tau_1 u_\chi|. \quad (6)$$

Moreover

$$Z'_m(0, A) = \frac{2}{|G|} \sum_{\chi(R)=-1} \chi(A)^{-1} L'_m(0, \chi). \quad (7)$$

Consequently a common denominator in (6)–(7) gives an explicit power identity for every X_A .

Remark 5. Here $w_K = w_{L_\chi} = 2$: both fields have a real embedding, so their only roots of unity are ± 1 . The root-of-unity quotient in (6) is retained to show its class-number-formula origin but is identically one in every application.

Proof. For a primitive quadratic χ° , $\zeta_{L_\chi}(s) = \zeta_K(s) L(s, \chi^\circ)$. The analytic class-number formula at zero, with the same regulator normalization as (4), is

$$\zeta_F(s) = -\frac{h_F R_F}{w_F} s^{r_1(F)+r_2(F)-1} + O\left(s^{r_1(F)+r_2(F)}\right).$$

Since the unit ranks of K and L_χ are one and two, respectively, division gives

$$L'(0, \chi^\circ) = \frac{h_{L_\chi}}{h_K} \frac{w_K}{w_{L_\chi}} \frac{R_{L_\chi}}{R_K}. \quad (8)$$

We next prove the regulator identity without suppressing the mixed signature. Put $a = \log |\tau_1 \epsilon_K|$, where τ_1, τ_2 lie over the split real place of K . The product formula gives

$$\lambda_L(\epsilon_K) = (a, a, -2a).$$

For the relative unit u_χ , exact norm one modulo torsion gives

$$\lambda_L(u_\chi) = (b, -b, 0), \quad b = \log |\tau_1 u_\chi| > 0.$$

The determinant of the first two coordinates is $2|a|b$. Since $R_K = |a|$, the regulator of $\langle E_K, u_\chi \rangle$ is $2R_K \log |\tau_1 u_\chi|$. The same subgroup has index I_χ in the full free unit group, so its regulator is $I_\chi R_{L_\chi}$. Therefore

$$\frac{R_{L_\chi}}{R_K} = \frac{2}{I_\chi} \log |\tau_1 u_\chi|. \quad (9)$$

If $\mathfrak{m}/\mathfrak{f}_{\chi^\circ}$ is the imprimitive part, then

$$L_{\mathfrak{m}}(s, \chi) = L(s, \chi^\circ) \prod_{\mathfrak{p}|\mathfrak{m}/\mathfrak{f}_{\chi^\circ}} (1 - \chi^\circ(\mathfrak{p})N\mathfrak{p}^{-s}).$$

At $s = 0$ the product is the exact integer E_χ . If one factor vanishes, both this product and $L(s, \chi^\circ)$ vanish at zero, so $L'_{\mathfrak{m}}(0, \chi) = 0$; otherwise combining (8) and (9) proves (6).

Finally, character inversion gives

$$\zeta_{\mathfrak{m}}(s, A) = \frac{1}{|G|} \sum_{\chi \in \hat{G}} \chi(A)^{-1} L_{\mathfrak{m}}(s, \chi).$$

Subtracting the formula with A replaced by RA multiplies the χ -term by $1 - \chi(R)^{-1}$, which is two precisely for $\chi(R) = -1$ and zero otherwise. Differentiation proves (7). Every surviving character is quadratic, hence every coefficient is rational. If q clears their denominators, exponentiation yields

$$X_A^q = \prod_{\chi(R)=-1} |\tau_1 u_\chi|^{n_{A,\chi}} \quad (n_{A,\chi} \in \mathbb{Z}), \quad (10)$$

after the exact factors in (6) have been absorbed into the integers $n_{A,\chi}$. This proves Proposition 4 and Theorem 1. $\square$

Corollary 6 (Closed packet formula). *Under the hypotheses of Theorem 1, put*

$$c_{A,\chi} = \frac{4}{|G|} \chi(A)^{-1} E_\chi \frac{h_{L_\chi}}{h_K} \frac{w_K}{w_{L_\chi}} \frac{1}{I_\chi}.$$

Then the positive real invariant at the selected embedding is

$$X_A = \prod_{\chi(R)=-1} |\tau_1 u_\chi|^{c_{A,\chi}}.$$

Every exponent $c_{A,\chi}$ is rational and is computed by exact finite arithmetic.

Remark 7. *The dominant per-character cost is one quartic-field call to `bnfinit`($P, 1$), certified by `bnfcertify`, followed by the 2×2 determinant in (5). Since χ° is quadratic, every imprimitive Euler factor at zero is 0 or 2, so E_χ is either 0 or 2^{r_χ} , where r_χ counts the nonvanishing factors. After the relative units are embedded in their compositum, the packet minimal polynomial is obtained by an exact resultant; no floating-point recognition enters this computation.*

The word “uniform” refers to this single character product, not to one field-independent named unit. For fixed $(K, \mathfrak{m}, A)$, the supported characters form a finite set, and substituting their exact class numbers, Euler factors, unit kernels, and determinant indices reduces the corollary immediately to a concrete product of explicitly computed relative units. If $E_\chi = 0$, that character contributes exponent zero and is simply absent from the product; if every supported character vanishes, the product is empty and $X_A = 1$.

The theorem is uniform; its proof does not enumerate the census. The corresponding queue statistics and exact Euler-degeneracy audit are reported in the supplement. The closed form is what makes the frozen quadratic-support queue exhaustively verifiable rather than sampled. In a mixed-support packet, its quadratic Fourier slice can be evaluated and removed exactly, so the certified-analysis budget of Engines B or C is spent only on the higher-order residual. Each application still certifies its primitive conductor, field, class numbers, Euler factors, exact norm map, primitive kernel, determinant, and orientation. The dimension-four and dimension-eight anchors independently replay (6), including relative index $I_\chi = 2$ in both dimension-eight cases.

2.1 A worked nonzero imprimitive row: RQ-000013

The following identity is **PROVED**: Proposition 4 is applied after every case-specific hypothesis is checked by exact arithmetic. The accompanying **bnrL1** point-value comparison in the certificate is only an **OBSERVED** cross-check and is not used in the proof.

Let $K = \mathbb{Q}(\sqrt{2})$, with finite ideal in the frozen integral basis

$$\mathfrak{f} = \begin{pmatrix} 14 & 6 \\ 0 & 2 \end{pmatrix}, \quad N\mathfrak{f} = 28,$$

and modulus $\mathfrak{m} = \mathfrak{f}\infty_2$. Its ray group is $G \simeq C_2$, in pinned PARI coordinates $G = \{[0], [1]\}$; the sign class is $R = [1]$. The unique supported character has $\chi([1]) = -1$. Its primitive conductor is

$$\mathfrak{f}_\chi = \begin{pmatrix} 7 & 3 \\ 0 & 1 \end{pmatrix} \infty_2.$$

The unique omitted prime has norm 2, valuation 2 in $\mathfrak{f}$, valuation 0 in $\mathfrak{f}_\chi$, and $\chi^\circ(\mathfrak{p}) = -1$. Hence the exact imprimitive factor is

$$E_\chi = 1 - \chi^\circ(\mathfrak{p}) = 2.$$

The quadratic ray field is represented relatively and absolutely by

$$t^2 - 2\sqrt{2} + 1, \quad P(t) = t^4 + 2t^2 - 7.$$

It has signature $(2, 1)$, discriminant -448 , class number 1, and two roots of unity; **bnfcertify** returns 1. On its free-unit basis the exact norm map is

$$\begin{pmatrix} 1 & -1 \end{pmatrix},$$

whose primitive kernel is generated by $(1, 1)^T$. The embedded base unit has coordinates $(1, -1)^T$, so

$$I_\chi = \left| \det \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \right| = 2.$$

Put

$$u = \frac{t^2 - 2t + 3}{4}.$$

The chosen absolute generator is the unique root of P in $(-34/25, -27/20)$. Exact Sturm counts and rational endpoint evaluation give

$$\frac{9}{5} < u < \frac{19}{10}.$$

Thus this is the oriented relative unit. It has minimal polynomial

$$U^4 - 2U^3 + U^2 - 2U + 1,$$

and the nontrivial Artin element $t \mapsto -t$ sends u exactly to u^{-1} .

Since $h_K = h_L = 1$, $w_K = w_L = 2$, $E_\chi = 2$, and $I_\chi = 2$, Proposition 4 gives

$$L'_\mathfrak{m}(0, \chi) = 2 \log u.$$

Here $|G| = 2$, so the Fourier coefficient $2/|G|$ is one. Therefore the fully Artin-labelled packet is

$$\boxed{X_{[0]} = u^2, \quad X_{[1]} = u^{-2}.}$$

The selected value $u^2 \in (7/2, 18/5)$ has minimal polynomial

$$X^4 - 2X^3 - 5X^2 - 2X + 1.$$

The exact source of truth is `artifacts/rq000013-engine-a-imprimitive-certificate-v1.json`; its replay command is `python3 scripts/certify_rq000013_engine_a.py --check`.

3 Shintani transfer and height rigidity

Let $H = H_{\mathfrak{m}}$ be the one-place ray class field of $\mathfrak{m} = \mathfrak{f}\infty_2$, and let $\mathcal{N}$ be its normal closure over $\mathbb{Q}$. Engine B first checks Shintani's conditions (0-3), (0-6), and (0-9): respectively, -1 is absent from the image of the totally positive units modulo $\mathfrak{f}$, 1 is absent from the residue classes of norm-negative units, and the required one-place splitting holds with

$$[H : H \cap \mathbb{Q}^{\text{ab}}] = 2. \quad (11)$$

The maximal absolutely abelian subfield is the fixed field of $[\text{Gal}(\mathcal{N}/\mathbb{Q}), \text{Gal}(\mathcal{N}/\mathbb{Q})]$. The imaginary quadratic field and transfer conductor below are then reconstructed a second time from their intrinsic ray class fields.

More precisely, an Engine-B transfer route consists of an imaginary quadratic subfield $k \subset \mathcal{N}$, a finite modulus $\mathfrak{c}$ of k , and a k -isomorphism

$$\iota : H_{\mathfrak{c}}(k) \xrightarrow{\sim} \mathcal{N}.$$

Here $\mathcal{N}/k$ is abelian, $\mathfrak{c} = \text{cond}(\mathcal{N}/k)$, and $H_{\mathfrak{c}}(k)$ denotes the corresponding ray class field. Thus “transfer conductor” below means this Artin conductor, not a conductor guessed from a discriminant. In every selected case it is computed by `rnfconductor` and the isomorphism is independently constructed by `bnrclassfield`; the two routes printed for a case must produce the same $\mathcal{N}$.

Proposition 8 (Shintani transfer in one-place notation). *Let $C = \text{Cl}_{\mathfrak{f}\infty_1\infty_2}(K)$, and suppose exact fiber identities identify Shintani's invariant $X_{\mathfrak{f}}(c, G)$ with X_A in (1). Assume Shintani's three conditions just restated, and let k and $\mathfrak{c}$ be respectively the imaginary quadratic field and finite transfer conductor occurring in his proof.*

For $\mathfrak{d} \mid \mathfrak{c}$, let

$$h(\mathfrak{d}) = |\text{Cl}_{\mathfrak{d}}(k)|, \quad w(\mathfrak{d}) = |\{\zeta \in \mu(k) : \zeta \equiv 1 \pmod{\mathfrak{d}}\}|,$$

put

$$n(\mathfrak{d}) = w(\mathfrak{d}) \frac{h(\mathfrak{c})}{h(\mathfrak{d})}, \quad (12)$$

and, for $\mathfrak{d} \neq \mathcal{O}_k$, let $f_{\mathfrak{d}}$ be the positive generator of $\mathfrak{d} \cap \mathbb{Z}$. If r clears the explicitly computed real-side distribution and induction denominators, then

$$\begin{aligned} m_{\mathcal{O}_k} &= 12h_k n(\mathcal{O}_k), \\ m_{\mathfrak{d}} &= 12f_{\mathfrak{d}} n(\mathfrak{d}) \quad (\mathfrak{d} \neq \mathcal{O}_k), \\ m &= \text{lcm}(r, (m_{\mathfrak{d}})_{\mathfrak{d} \mid \mathfrak{c}}) \end{aligned} \quad (13)$$

has the following property:

$$X_A^m \in \mathcal{O}_H^{\times}, \quad (X_A^m)^{\text{Art}(B)} = X_{AB}^m. \quad (14)$$

At every embedding of H above the nonsplit real place, $|X_A^m| = 1$.

Proof. Here and in this paragraph, unqualified theorem, proposition, and equation numbers refer to Shintani's paper [3]. His definition (0-5) on p. 140 introduces the positive invariant $X_{\mathfrak{f}}(c)$, and Theorem 1 on p. 140 identifies its logarithm with the difference of the two partial zeta derivatives. Definition (18) on p. 150 forms the distribution-corrected invariant $Y_{\mathfrak{f}}(c)$; definitions (19)–(20) on pp. 151–152 then take the products over G . The assumed exact fiber identity identifies our X_A with $X_{\mathfrak{f}}(c, G)$, while the explicit real-side distribution calculation, whose denominators divide r , passes from $X_{\mathfrak{f}}(c, G)$ to $Y_{\mathfrak{f}}(c, G)$. Under these three hypotheses, Shintani's Theorem 2 on p. 141 supplies the index-two algebraicity construction.

On the imaginary-quadratic side, Shintani's equation (8) on p. 146 defines

$$n(S) = w(\mathfrak{f}(S)) \frac{|H_k(\mathfrak{c})|}{|H_k(\mathfrak{f}(S))|}.$$

Reindexing the subsets by the divisors $\mathfrak{d} = \mathfrak{f}(S)$ of $\mathfrak{c}$ gives exactly (12). For $\mathfrak{d} \neq \mathcal{O}_k$, Shintani's equation (4) on p. 143 contains $|\varphi_{\mathfrak{d}}|^{1/(6f_{\mathfrak{d}})}$, and equation (9) on p. 146 contributes the further exponent $1/n(\mathfrak{d})$. Replacing an absolute value by its product with its complex conjugate contributes a factor two. Hence $12f_{\mathfrak{d}}n(\mathfrak{d})$ clears that row. At the trivial divisor, equation (6) on p. 144 replaces $f_{\mathfrak{d}}$ by h_k , giving $12h_kn(\mathcal{O}_k)$.

Shintani's Proposition 4 on pp. 154–156 expresses $Y_{\mathfrak{f}}(c, G)$ as an integral product of ratios of the $W_{\mathfrak{c}}$ -invariants and introduces no additional rational power. The integer r in (13), which is external to Shintani's imaginary-side construction, clears the explicitly recorded real-side fiber, distribution, and induction denominators. Thus (13) clears every exponent in his product. Finally, Shintani's Proposition 5(i)–(iii) on pp. 156–158 supplies, respectively, the unit and field assertion, Artin covariance, and modulus one under embeddings inducing the nontrivial real automorphism. This proves (14). $\square$

For the headline case $K = \mathbb{Q}(\sqrt{7})$, $\mathfrak{f} = \begin{bmatrix} 7 & 0 \\ 0 & 1 \end{bmatrix}$, the short transfer route has $k = \mathbb{Q}(i)$ and $\mathfrak{c} = (7)$. The complete divisor table is

$\mathfrak{d}$	$h(\mathfrak{d})$	$w(\mathfrak{d})$	$n(\mathfrak{d})$	$f_{\mathfrak{d}}$	$m_{\mathfrak{d}}$
$\mathcal{O}_k$	1	4	48	1	576
(7)	12	1	1	7	84

(15)

and the real-side indices are 6 and 1. Hence $\text{lcm}(576, 84, 6) = 4032$. This is a complete instance of Proposition 8; the longer $\mathbb{Q}(\sqrt{-7})$ reconstruction gives the same normal closure and safe power independently.

Lemma 9 (All-embeddings height rigidity). *Let H be a number field of degree D . Suppose $(X_A^m)_A \subset \mathcal{O}_H^\times$ is an algebraic family with Artin covariance, and let $(\alpha_A)_A \subset \mathcal{O}_H^\times$ be an exactly Artin-labeled positive candidate family. For each archimedean place v of H , choose an embedding $\sigma_v : H \hookrightarrow \mathbb{C}$ representing v , and suppose either*

$$\left| \frac{1}{m} \log |\sigma_v(X_A^m)| - \log |\sigma_v(\alpha_A)| \right| \leq \epsilon_v \quad (16)$$

or both powered absolute values are proved to be one. Put

$$B_A = \frac{m}{D} \sum_{v|\infty} n_v \epsilon_v, \quad n_v = [H_v : \mathbb{R}], \quad (17)$$

with $\epsilon_v = 0$ in the second case. If

$$B_A < \min \left\{ \frac{1}{2} \log \frac{1 + \sqrt{5}}{2}, \min_{3 \leq d \leq D} \frac{1}{4d} \left(\frac{\log \log d}{\log d} \right)^3 \right\}, \quad (18)$$

then $X_A = \alpha_A$.

Proof. The quotient $\eta_A = X_A^m / \alpha_A^m$ is a unit of H . Its finite-place contribution to the Weil height is zero. Equations (16)–(17) and the definition of logarithmic height give

$$h(\eta_A) = \frac{1}{D} \sum_{v|\infty} n_v \max(0, \log |\eta_A|_v) \leq B_A.$$

If $d = [\mathbb{Q}(\eta_A) : \mathbb{Q}] \geq 3$ and η_A is not torsion, the main theorem of Voutier [9, p. 82], stated for every degree $d \geq 2$, contradicts (18). A non-torsion quadratic unit has height at least $\frac{1}{2} \log((1 + \sqrt{5})/2)$, and a rational unit is ± 1 . Thus η_A is a root of unity. At the distinguished real embedding the analytic number X_A and the algebraic candidate α_A are positive. Hence $\eta_A = 1$, and $X_A^m = \alpha_A^m$ there; taking the unique positive m -th root proves the claim. $\square$

For every row of Theorem 2, the embeddings over the split real place are all represented by the printed packet roots and the cone enclosures. At the remaining archimedean places, Proposition 8 gives modulus one for X_A^m , and the reciprocal candidate polynomial is checked exactly, after the substitution $T = X + X^{-1}$, to have its remaining roots on the unit circle. The exact ray-field isomorphism pairs each such candidate root with the same nonsplit place at which Proposition 8 gives $|\sigma_v(X_A^m)| = 1$; hence $|\sigma_v(\alpha_A)| = 1$ at that specific place, and it contributes $\epsilon_v = 0$ in (17).

The algebraic candidates are certified without a GRH assumption: each class and unit computation uses `bnfinit(P,1)`, followed by the full call `bnfcertify(bnf)` returning 1; the quotient-only flag is not used. Exact `rnfconductor` and `bnrclassfield` round trips identify the relative ray fields, Sturm sequences isolate the positive roots, and split-prime Frobenius actions label them.

Effective but not uniformly cheap. The required raw radius is $O(V_{\min}/m)$. Arb precision therefore grows with $\log m$, but conductor size, packet degree, and cone count also grow, and no uniform complexity claim is made. The order-ten case below has $m = 15840$ and remains comfortable; cases with much larger exponents can exceed the fixed per-case resource cap and belong to the frontier. This is a stated limit of Engine B, not evidence against its completed cases.

4 The order-six and order-ten packets

4.1 $\mathbb{Q}(\sqrt{7})$ at the ramified prime above seven

For $K = \mathbb{Q}(\sqrt{7})$ and $\mathfrak{m} = \mathfrak{p}_{7\infty_2}$, the one-place and two-place ray groups are C_6 and $C_6 \times C_2$. The support orders are two and six and (11) holds. The commutator-fixed calculation produces $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-7})$, and their intrinsic conductor reconstructions give the same degree-24 normal closure. The $\mathbb{Q}(i)$ route has two divisor rows and safe exponent 4032.

The packet polynomial over K is

$$\begin{aligned} P_7(X) = & X^6 - (6 + 2\sqrt{7})X^5 + (22 + 8\sqrt{7})X^4 \\ & - (34 + 13\sqrt{7})X^3 + (22 + 8\sqrt{7})X^2 \\ & - (6 + 2\sqrt{7})X + 1. \end{aligned} \tag{19}$$

It is irreducible over K , defines the one-place ray field, has six positive Sturm-isolated roots, and is Artin-labeled by the split primes 19 and 31.

For the class represented by $(3)^r$, an exact Yamamoto domain has generators

$$\lambda_r = (7 - 2\sqrt{7})/3^r, \quad \lambda_r(8 + 3\sqrt{7}).$$

Its determinant is nine and its half-open parallelotope contains exactly nine lattice points. The resulting 27 independent double-sine evaluations are enclosed without a PARI L -value. At exponent 4032, the powered height upper bound is 9.1903×10^{-9} . The minimum Voutier lower bound in degrees three through 24 is 5.2279×10^{-5} , so the margin exceeds 5688. The degree-one and degree-two fallbacks finish the proof of (19).

This is the headline order-six instance. The same character order is the difficult component at the separate $\mathbb{Q}(\sqrt{21})$ wall, but here the index-two hypothesis holds and the packet closes. Thus the character order itself is not the obstruction.

4.2 Replications and the ramified-prime-3 control

For $\mathbb{Q}(\sqrt{14})$ at $\mathfrak{p}_7$, both $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{-2})$ reconstruct the degree-24 normal closure. The latter route has clearing exponents 576 and 84, with least common multiple 4032. The packet is

$$\begin{aligned} P_{14}(X) = & X^6 - (13 + 4\sqrt{14})X^5 + (85 + 22\sqrt{14})X^4 \\ & - (139 + 38\sqrt{14})X^3 + (85 + 22\sqrt{14})X^2 \\ & - (13 + 4\sqrt{14})X + 1. \end{aligned} \quad (20)$$

Split primes 11 and 103 label the Artin generators. The 20-point cone evaluator gives powered height at most 7.1466×10^{-9} , hence margin greater than 7315.

The strongest boundary control is RQ-002057 over $\mathbb{Q}(\sqrt{57})$, with modulus of norm 27. The conductor is a power of the ramified prime 3 and the support still has order six. The bases $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{-19})$ independently give the same degree-24 closure. On the shorter route the divisor exponents are 864, 324, 108, hence $m = 2592$. With $y = (1 + \sqrt{57})/2$, so $y^2 - y - 14 = 0$, its packet is

$$\begin{aligned} P_{57}(X) = & X^6 - (30 + 9y)X^5 + (582 + 177y)X^4 \\ & - (2529 + 772y)X^3 + (582 + 177y)X^2 \\ & - (30 + 9y)X + 1. \end{aligned} \quad (21)$$

The fundamental positive unit has order three modulo the finite modulus, so a ray-class domain is the union of three adjacent cones, with 240 exact affine points per class. The powered height is at most 6.982×10^{-8} , giving margin greater than 748. Thus order-six support and ramified-prime-3 structure are both reachable; neither alone explains the separate $\mathbb{Q}(\sqrt{21})$ boundary.

4.3 Three further Shintani closures

The three remaining rows of Theorem 2 use exactly the Engine-B proof above; we print their routes and algebraic candidates so that no theorem row is supported only by a corpus identifier.

For RQ-000108, $K = \mathbb{Q}(\sqrt{5})$ and $N\mathfrak{f} = 45$. The two imaginary bases are $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{-15})$, and both reconstruct the same degree-16 normal closure. With $y = (1 + \sqrt{5})/2$, the relative packet polynomial is

$$X^4 - (1 + 6y)X^3 + (9 + 9y)X^2 - (1 + 6y)X + 1. \quad (22)$$

It is the certified one-place ray field over K . At $m = 2880$, the powered height is at most 2.12465×10^{-8} , so the Voutier margin exceeds 2460.

For RQ-000021, $K = \mathbb{Q}(\sqrt{2})$ and $\mathfrak{f} = (7)$. The bases $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{-14})$ reconstruct the same degree-24 closure. The packet polynomial is

$$\begin{aligned} & X^6 - (10 + 6\sqrt{2})X^5 + (58 + 40\sqrt{2})X^4 \\ & - (129 + 90\sqrt{2})X^3 + (58 + 40\sqrt{2})X^2 \\ & - (10 + 6\sqrt{2})X + 1. \end{aligned} \quad (23)$$

The cone boundary meets a split-prime face; the canonical value is the average of the upper and lower half-open conventions. Both are enclosed independently. At $m = 2016$, the powered height is at most 1.22671×10^{-8} , giving margin greater than 4261.

For RQ-002955, $K = \mathbb{Q}(\sqrt{77})$ and $\mathfrak{f} = \mathfrak{p}_7$ has HNF $\begin{bmatrix} 7 & 3 \\ 0 & 1 \end{bmatrix}$. The bases $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{-11})$ give the same degree-24 closure. Put $y = (1 + \sqrt{77})/2$, so $y^2 - y - 19 = 0$. The packet is

$$\begin{aligned} & X^6 - (15 + 3y)X^5 + (107 + 26y)X^4 \\ & - (217 + 54y)X^3 + (107 + 26y)X^2 - (15 + 3y)X + 1. \end{aligned} \quad (24)$$

At $m = 4032$, the powered height is at most 1.01485×10^{-8} , and the margin exceeds 5151. In all three cases the exact ray-field identification, Sturm isolation, Artin labels, and the degree-one and degree-two fallbacks are included in the cited case records.

4.4 Order ten

For RQ-001107 over $\mathbb{Q}(\sqrt{33})$ and $\mathfrak{p}_{11}$, put $y = (1 + \sqrt{33})/2 > 0$, so $y^2 - y - 8 = 0$. The packet is

$$\begin{aligned} P_{33}(X) = & X^{10} - (8 + 4y)X^9 + (74 + 30y)X^8 - (294 + 125y)X^7 \\ & + (669 + 281y)X^6 - (871 + 368y)X^5 + (669 + 281y)X^4 \\ & - (294 + 125y)X^3 + (74 + 30y)X^2 - (8 + 4y)X + 1. \end{aligned} \quad (25)$$

Its relative degree is ten, so the certified ray field H/K has absolute degree 20. The normal closure used to compute the Frobenius action has degree 40, but the comparison quotient $\eta_A = X_A^m / \alpha_A^m$ lies in H , since both powered terms do. Therefore Lemma 9 needs only $d \leq 20$. The archived computation conservatively evaluated every degree through 80; on either window the minimum Voutier bound is

$$5.22795332222684 \dots \times 10^{-5}.$$

At $m = 15840$, the factor-100 raw-error target is $3.30047558221391 \dots \times 10^{-11}$; the achieved bound is 5.6731×10^{-13} , so the certified margin exceeds 5817. The degree-one and degree-two fallbacks are recorded separately. This support order is the first one in the present proved inventory divisible by the prime 5; its historical priority status is the survey-bounded status stated after Theorem 2.

The seven Engine-B rows treated in this section, together with the RQ-000458 Engine-B proof in Section 6.4, complete the proof of Theorem 2.

5 Cyclic-quartic CM descent

Let $G = \text{Cl}_m(K)$, let θ be a supported quartic character, and put $U = \ker \theta$. Fix $\bar{s} \in G/U$ with $\theta(\bar{s}) = i$. By character inversion and the definition (2),

$$\log Y_{\bar{s}^r} = \frac{1}{2} \left(i^{-r} L'_m(0, \theta) + i^r L'_m(0, \bar{\theta}) \right) = \Re(i^{-r} L'_m(0, \theta)). \quad (26)$$

Indeed averaging over U kills every character nontrivial on U ; the only odd characters of $G/U \simeq C_4$ are $\theta, \bar{\theta}$. This is the missing link between an individual one-place packet component and a quartic L' -value.

For Engine C, let k be imaginary quadratic and let E/k be a cyclic quartic extension with

$$C = \text{Gal}(E/k) = \langle \sigma \rangle, \quad \psi(\sigma) = i.$$

Write $\tau = \sigma^2$ for the internal involution over k . Global complex conjugation is a different automorphism, denoted by j , of the common normal closure, because it acts nontrivially on the imaginary quadratic base.

Theorem 10 (Cyclic-quartic CM norm bridge). *Use the Fourier, character, and Artin conventions fixed in Section 1. Assume the following exact hypotheses.*

1. *Linear reinduction, including the imprimitive local factors, gives*

$$\text{Ind}_{G_K}^{G_{\mathbb{Q}}} \theta \simeq \text{Ind}_{G_k}^{G_{\mathbb{Q}}} \psi, \quad L_m(s, \theta) = L_S(s, \psi). \quad (27)$$

2. Stark's proved imaginary-quadratic rank-one theorem applies to $(E/k, S)$, with $|S| \geq 3$, and supplies a global unit ε , unique modulo $\mu(E)$. Put $e = |\mu(E)|$.
3. In $\mathcal{O}_E^\times/\mu(E)$, $\tau(\varepsilon) = \varepsilon^{-1}$ and $\varepsilon = u^{e/2}$ for a certified free unit u .
4. The exact normal-closure embedding identifies $\bar{s}$ with σ , and global complex conjugation satisfies

$$j(E) = E, \quad j|_k \neq 1.$$

Thus $j \in \text{Aut}(E)$, and the embedding identifies this automorphism with complex conjugation on E .

The divisibility assertion in item 3 is certified directly in the imaginary-quadratic unit lattice. It is not an application of Roblot's squareness criteria [13, Cor. 2.4 and Thms. 6.2, 7.2], whose framework assumes a totally real base and explicitly excludes imaginary quadratic bases. Put $E^+ = E^{(j)}$. Then E/E^+ is quadratic and the norm below is defined. For $r \in \mathbb{Z}/4\mathbb{Z}$,

$$\boxed{Y_{\bar{s}^r} = N_{E/E^+}(\sigma^r u)^{-1}.} \quad (28)$$

The right side is independent of the root-of-unity representative of ε . Thus exact character selection, unit divisibility, and the common-normal-closure identities identify the entire Artin-labeled primitive quartic packet.

Proof. Stark's normalized absolute value at the complex place is the square of the ordinary complex modulus. If $\ell_g = \log |g\varepsilon|_{\text{ord}}$, his proved formula therefore reads

$$\zeta'_S(0, g) = -\frac{2}{e}\ell_g. \quad (29)$$

The anti-unit relation gives $\ell_{\sigma^2} = -\ell_1$ and $\ell_{\sigma^3} = -\ell_\sigma$. Fourier transformation with $\psi(\sigma) = i$ gives

$$L'_S(0, \psi) = -\frac{4}{e}(\ell_1 + i\ell_\sigma). \quad (30)$$

These are respectively the class-log coefficient $2/e$ and the direct- L' coefficient $4/e$; their inverse recovery coefficients are $-e/2$ and $-e/4$, with the same plus sign in the latter Gaussian combination.

Write $m_r = \log |\sigma^r u|_{\text{ord}}$. The exact divisibility in hypothesis 3 gives $\ell_{\sigma^r} = (e/2)m_r$, so (30) becomes

$$L'_S(0, \psi) = -2(m_0 + im_1).$$

By (27) this is $L'_m(0, \theta)$. Substitution in (26) yields, successively,

$$(\log Y_1, \log Y_{\bar{s}}, \log Y_{\bar{s}^2}, \log Y_{\bar{s}^3}) = (-2m_0, -2m_1, 2m_0, 2m_1).$$

Since the internal anti-unit relation gives $(m_2, m_3) = (-m_0, -m_1)$, while

$$\log N_{E/E^+}(\sigma^r u) = 2m_r$$

at the selected complex embedding, this vector is exactly the logarithm of the right side of (28). Both sides are positive, so equality of logarithms proves (28). If u is multiplied by a root of unity ξ , then $j(\xi) = \xi^{-1}$, and its CM norm is unchanged.

The positive power σ^r is the convention checked by the Artin-label replay: its exact bridge applies the selected lift of σ exactly r times before taking the CM norm. Thus the replay uses the same generator as $\theta(\bar{s}) = \psi(\sigma) = i$, rather than silently inverting one side. $\square$

Finite character selection and algebraic closure. The hypotheses of Theorem 10 are finite checks. The pipeline enumerates every linear ray character compatible with the relative conductor and the common normal closure. Scalar twists are not permitted: (27) is checked as an equality of ordinary characters on every conjugacy class. Exact Dirichlet coefficients are appended until they separate the finite candidate set. The selected L' -value is enclosed by an independent theta/Mellin evaluation, and Arb inversion in $\mathcal{O}_E^\times/\mu(E)$ must isolate one integral anti-unit orbit. Exact norm identities in the common normal closure then verify (28) at every Artin label. Equality of unlabeled polynomials is never the bridge.

6 CM packet identifications

6.1 The mixed-class-number packet over $\mathbb{Q}(\sqrt{35})$

For RQ-001280 the primitive finite ideal has HNF $\begin{bmatrix} 8 & 4 \\ 0 & 4 \end{bmatrix}$ and norm 32. The two exact routes use $\mathbb{Q}(\sqrt{-10})$ and $\mathbb{Q}(\sqrt{-14})$. Their character fields have $e = 2$, their base class numbers are respectively 2 and 4, and $|S| = 3$ in both routes. The exact coefficient selector, Arb unit-orbit isolation, and common-normal-closure identities give

$$\begin{aligned} P_{35}(X) = & X^8 - 38904X^7 + 905404X^6 + 62136X^5 \\ & - 873210X^4 + 62136X^3 + 905404X^2 - 38904X + 1. \end{aligned} \quad (31)$$

It has four positive roots and two nonreal conjugate pairs. Thus the four real roots are precisely the packet (28), while the full selector field has signature $[4, 2]$. Exact conductor transport also proves the member at HNF $\begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix}$ of norm 64.

6.2 The $e = 6$ tranche

The following three cases each have two independent cyclic-quartic routes, $|S| \geq 3$, and the exact divisibility $\varepsilon = u^3$ modulo $\mu(E)$. Theorem 10 therefore applies without a new normalization.

For RQ-001569 over $\mathbb{Q}(\sqrt{42})$, at HNF $\begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}$, the bases are $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{-6})$, and

$$\begin{aligned} P_{42}(X) = & X^8 - 416X^7 + 2332X^6 - 2144X^5 \\ & + 2758X^4 - 2144X^3 + 2332X^2 - 416X + 1. \end{aligned} \quad (32)$$

For RQ-007519 over $\mathbb{Q}(\sqrt{186})$, at the same HNF, the bases are $\mathbb{Q}(\sqrt{-6})$ and $\mathbb{Q}(\sqrt{-31})$, and

$$\begin{aligned} P_{186}(X) = & X^8 - 90243296X^7 + 206015683996X^6 \\ & + 697017342688X^5 + 984190237126X^4 \\ & + 697017342688X^3 + 206015683996X^2 - 90243296X + 1. \end{aligned} \quad (33)$$

For RQ-001894 over $\mathbb{Q}(\sqrt{51})$, at HNF $\begin{bmatrix} 15 & 0 \\ 0 & 5 \end{bmatrix}$, the bases are $\mathbb{Q}(\sqrt{-15})$ and $\mathbb{Q}(\sqrt{-85})$, and

$$\begin{aligned} P_{51}(X) = & X^8 - 1430858X^7 + 316969003X^6 - 224768756X^5 \\ & + 477153445X^4 - 224768756X^3 + 316969003X^2 \\ & - 1430858X + 1. \end{aligned} \quad (34)$$

Each polynomial has four positive roots and two nonreal conjugate pairs, and each pair of routes gives identical Artin labels, not just the same polynomial. These three polynomials supersede the larger powered polynomials in the first $e = 6$ bridge record: that record took CM norms of ε , whereas (28) requires the exact cube root u . The correction divides the certified free-unit coordinates by three and reruns both normal-closure routes.

6.3 $\mathbb{Q}(\sqrt{6})$: one proof and one quarantined cross-check

For RQ-000129 the finite ideal has HNF $\begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$. The route over $\mathbb{Q}(\sqrt{-2})$ has $e = 8$ and its natural Stark set already has $|S| = 3$. The exact unit lattice gives $\varepsilon = u^4$ modulo $\mu(E)$. Thus Theorem 10 directly proves

$$P_6(X) = X^8 - 8X^7 + 12X^6 + 8X^5 - 10X^4 + 8X^3 + 12X^2 - 8X + 1. \quad (35)$$

Sturm isolation gives four positive roots and no negative roots.

The second route over $\mathbb{Q}(\sqrt{-3})$ has $e = 12$, but its natural set has $|S| = 2$. Enlargements at the rational primes 3 and 5 reproduce the same analytic orbit and the exact identity $q_8^3 = q_{12}^2$. These facts are a strong cross-check, but they are not used in the theorem: the current record does not print the individual auxiliary prime ideals and finite valuations needed to exclude a residual S -unit ambiguity. The earlier zero-real-root polynomial belonged to the complex anti-unit before unit-lattice division and CM norm; it is not a packet polynomial.

6.4 An Engine-B packet and a CM normalization diagnostic

For RQ-000458, $K = \mathbb{Q}(\sqrt{14})$, the finite ideal has HNF $\begin{bmatrix} 12 & 0 \\ 0 & 6 \end{bmatrix}$, the ray group is $C_4 \times C_2$, and the support is the inverse quartic pair. The Engine-B proof identifies

$$P_{458}(X) = X^4 - (20 + 6\sqrt{14})X^3 + (138 + 36\sqrt{14})X^2 - (20 + 6\sqrt{14})X + 1. \quad (36)$$

The safe exponent is 1152, and its powered height bound 8.0791×10^{-9} gives margin > 6470 . The Engine-C character calculation over $\mathbb{Q}(\sqrt{-42})$ and $\mathbb{Q}(\sqrt{-3})$ independently selects the same quartic L -function, but the banked orientation record uses a fourth-root normalization rather than the $e/2$ -root normalization in Theorem 10. Its isolated orbit is

$$(-8, 4), \quad (0, -4), \quad (8, -4), \quad (0, 4),$$

and 32 exact common-closure identities recover (36), but that calculation is retained here only as a diagnostic until its Fourier-scaling convention is rederived. The theorem claim for this row rests solely on Engine B.

7 Structural boundary lemmas

Lemma 11 (Absolutely abelian ray fields). *Let H/K be the one-place ray class field of a real quadratic field. If $H/\mathbb{Q}$ is Galois and abelian, then the sign class $R \in \text{Gal}(H/K)$ is trivial and the differenced Fourier support in (1) is empty.*

Proof. If the one-place field $H/\mathbb{Q}$ were abelian, the complex conjugations above the two real places of K would be conjugate: any lift of the nontrivial element of $\text{Gal}(K/\mathbb{Q})$ interchanges the places. In an abelian group conjugate elements are equal. The inertia element at the omitted real place is trivial in the one-place ray field, whereas the inertia element at the retained place is R . Hence $R = 1$. It follows immediately that no character satisfies $\chi(R) = -1$, so the differenced Fourier support is empty. $\square$

Engine A is not an exception: Proposition 4 reduces individual quadratic characters through (6)–(7), but it does not assert that the entire mixed-signature one-place ray field is absolutely abelian.

Lemma 12 (Index parity). *Let H/K be a one-place ray field and let $N/\mathbb{Q}$ be its genuine normal closure. If the sign class R is nontrivial, then*

$$2 \mid [H : H \cap \mathbb{Q}^{\text{ab}}]. \quad (37)$$

Equivalently, an odd Shintani index can occur only when the Fourier support of (1) is empty.

Proof. Put $G = \text{Gal}(N/\mathbb{Q})$, $A = \text{Gal}(N/K)$, and $B = \text{Gal}(N/H)$. Let $c_1, c_2 \in A$ be the real-place inertia involutions. A lift of the nontrivial automorphism of $K/\mathbb{Q}$ interchanges them, so they are conjugate in G . Hence $c_1 c_2^{-1} \in [G, G]$. Since $G/A \simeq C_2$, one has $[G, G] \subseteq A$. Moreover B is normal in A , so $[G, G]B$ is a subgroup of A . In $A/B = \text{Gal}(H/K)$, the involution at the omitted place is trivial and the one at the retained place is R . Thus the image of $[G, G]$ in A/B contains R . Its order is the degree of H over its fixed field. That fixed field is

$$H \cap N^{[G, G]} = H \cap \mathbb{Q}^{\text{ab}},$$

because $N^{[G, G]}$ is the maximal abelian subextension of $N/\mathbb{Q}$. Hence the image has order $[H : H \cap \mathbb{Q}^{\text{ab}}]$, which is even if $R \neq 1$. Finally, characters of a finite abelian group separate a nonidentity element of order two, so $R = 1$ is equivalent to empty differenced support. $\square$

The use of the genuine normal closure in Lemma 12 is essential. Conjugation coordinates at an unstable modulus do not define $[G, G]$. As an independent consistency check, a pre-registered replay over all 8,200 genuine normal-closure reconstructions found 446 odd indices greater than one. Every one had trivial sign class and empty differenced Fourier support; there were no exceptions.

A Exact finite moduli

The short labels in Theorems 2 and 3 are local to their rows. In the integral basis $1, \omega$ fixed in Theorem 2, the exact finite ideals are as follows.

case	short label	HNF(f)
RQ-000190	$\mathfrak{p}_7$	$\begin{pmatrix} 7 & 0 \\ 0 & 1 \end{pmatrix}$
RQ-000419	$\mathfrak{p}_7$	$\begin{pmatrix} 7 & 0 \\ 0 & 1 \end{pmatrix}$
RQ-000108	$\mathfrak{f}_{45}$	$\begin{pmatrix} 15 & 6 \\ 0 & 3 \end{pmatrix}$
RQ-000021	(7)	$\begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix}$
RQ-002057	$\mathfrak{f}_{27}$	$\begin{pmatrix} 9 & 3 \\ 0 & 3 \end{pmatrix}$
RQ-002955	$\mathfrak{p}_7$	$\begin{pmatrix} 7 & 3 \\ 0 & 1 \end{pmatrix}$
RQ-001107	$\mathfrak{p}_{11}$	$\begin{pmatrix} 11 & 5 \\ 0 & 1 \end{pmatrix}$
RQ-000458	$\mathfrak{f}_{72}$	$\begin{pmatrix} 12 & 0 \\ 0 & 6 \end{pmatrix}$
RQ-001280	$\mathfrak{f}_{32}$	$\begin{pmatrix} 8 & 4 \\ 0 & 4 \end{pmatrix}$
RQ-001569	$\mathfrak{f}_{12}$	$\begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$
RQ-007519	$\mathfrak{f}_{12}$	$\begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$
RQ-001894	$\mathfrak{f}_{75}$	$\begin{pmatrix} 15 & 0 \\ 0 & 5 \end{pmatrix}$
RQ-000129	$\mathfrak{f}_8$	$\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$

B Certificates and data availability

The exact field computations use PARI/GP 2.15.4 [16]; analytic certificates use Python 3.12.3 and python-flint 0.9.0 [17]. The principal replays were run under Linux/x86-64 on an AMD EPYC 9354P virtual CPU. All analytic conclusions are Arb balls, never point estimates. AI-assisted systems proposed code, text, and adversarial checks but were not mathematical authorities: every accepted computation was rerun by the pinned exact or Arb pipeline and reviewed by the author. The disclosure immediately before the bibliography gives the tools and purposes.

The certificate archive has three layers: exact field and ray-class records, Arb enclosure transcripts, and compact replay programs that check the theorem-facing identities against both. Supplementary Table S1 gives the complete record map. Supplementary Table S2 gives the complete Artin-label interval replay. For illustration, the headline case has

case	r : isolating interval
RQ-000190	0 : (5.325, 5.326), 1 : (2.713, 2.714), 2 : (2.251, 2.252),
	3 : (.187, .188), 4 : (.368, .369), 5 : (.444, .445)

The cone evaluation for class r , Shintani’s covariance (14), and the exact ray-field isomorphism identify the isolated root with X_r .

Archive DOI. The paper, supplementary tables, and frozen companion archive are deposited at <https://doi.org/10.5281/zenodo.21713178>. The PDF and $\text{\TeX}$ source are exposed as top-level files in that record. The companion archive contains the manuscript source, a complete SHA-256 manifest, expected verifier outputs, exact and Arb certificates, and a short README separating discovery programs from proof-checking programs. Separate minimal verification commands will cover the quadratic theorem, every selected Engine-B packet, the cyclic-quartic bridge and selected Engine-C packets, and the two structural lemmas. The environment record will state the software versions, PARI proof modes (including `bnfinit(P,1)` followed by `bnfcertify(bnf)=1`), and expected runtime and memory.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this work, the author used OpenAI GPT-5.6 for mathematical brainstorming, symbolic-computation scaffolding, code review, and manuscript editing, and Anthropic Claude for independent adversarial checking of statements and certificates. The author reviewed and verified all outputs, made all mathematical and editorial decisions, and takes full responsibility for the content of the article.

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